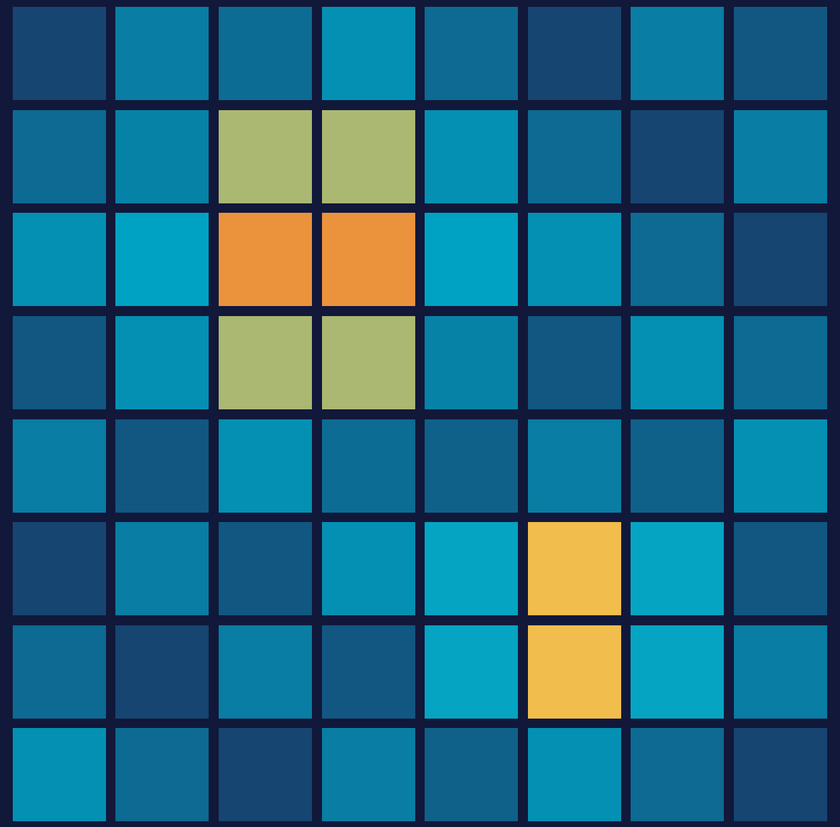
SATELLITE IMAGING

Satlas-SR

Uncertainty Calibration & Guided Reconstruction

Making multi-temporal satellite super-resolution more reliable through spatial–temporal uncertainty and adaptive, uncertainty-guided reconstruction.

Sentinel-2 10 m → 4× reconstruction → 2.5 m GSD



Per-pixel uncertainty map

THE PROBLEM

A sharp output can still be unreliable

A deterministic model returns one image, with no signal about which details may be wrong.

2–5 days

Sentinel-2(satellite image dataset) revisit, free and global, 10 m per pixel (Input images)

0.6–1 m

USDA NAIP airborne imagery: sharp, but costly and slow to refresh (ground truth image)

Blind hallucination

Cloud shadows, registration jitter and seasonal change produce plausible but false texture.

Perception–distortion dilemma

A single frame keeps haze and noise; uniform averaging cleans noise but blurs real edges.

Downstream risk

Analysts and vision models may treat reconstructed detail as observed evidence.

THE CORE IDEA

From sharp images to trustworthy reconstructions

Estimate where the model may be wrong, then blend predictions only as much as needed.

INPUT

8 Sentinel-2 frames

Temporal stack at 10 m, within ±3 months of the NAIP flight

BASE MODEL

23-block RRDBNet

Frozen 4× generator with 16.7 M parameters

UNCERTAINTY

Spatial + temporal

Calibrated per-pixel map built from two independent signals

RECONSTRUCTION

Soft blending

Sigmoid weights mix the primary and ensemble predictions

RESULT PREVIEW

locked holdout, 107 scenes

+39.5%

error-rank correlation (Sobel gradient)

+0.571 dB

PSNR over baseline (Sobel gradient)

0 retraining

frozen weights

APPROACH

A three-phase, audit-controlled workflow

Experiments progress from measurement integrity to uncertainty, then to guided reconstruction.

1 **EXP-001 – 013**

Integrity & baseline

- Score the 120×120 core window
- Audit tiles; freeze 107/107 split

2 **EXP-014 – 019**

Uncertainty calibration

- Spatial D4 test-time augmentation
- Temporal input sensitivity
- Hybrid map; one holdout test

3 **EXP-020A – 024**

Guided reconstruction

- Hard top-K replacement, audited
- Spatial 5-fold robustness
- Soft blending; final holdout

SAFEGUARD Weights frozen · parameters tuned on development data only · holdout opened twice

DATA ORIGIN

Two sensors, one aligned 4× learning problem

Sentinel-2 imagery paired with airborne NAIP ground truth across the contiguous United States, 2019–2020.

Sentinel-2 L1C (input)

- 10 m per pixel, RGB true-colour tiles
- ESA Copernicus; revisit every 2–5 days
- Top-of-atmosphere reflectance, 8-bit

4×

USDA NAIP (target)

- USDA NAIP provides the ground-truth images at 2.5 m GSD, aligned with the model's 4× super-resolution output from Sentinel-2 imagery at 10 m GSD.
- Ground truth for loss and evaluation

1.1 M

urban training pairs within 5 km of US cities (population ≥ 50k)

$32^2 \rightarrow 128^2$

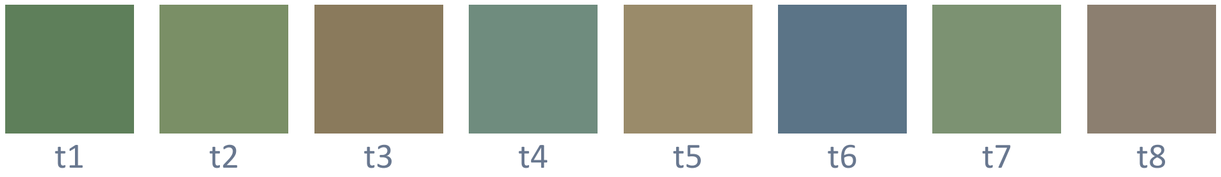
pixels per tile; the same 320 m × 320 m footprint

256

urban validation tiles used for standardized evaluation

Eight acquisitions become one 24-channel input

Multi-image super-resolution: the network sees the whole temporal stack at once.



each frame: [3, 32, 32] (RGB, 10 m)

concatenate along channels

$$X_{in} \in \mathbb{R}^{B \times 24 \times 32 \times 32}$$

channels 0–2 = frame 1 (R, G, B) … 21–23 = frame 8

1

Temporal window

Scenes limited to ±3 months of the NAIP flight date.

2

Quality filter

Tiles with no data edges, missing sensors or extreme shadow are culled.

3

Order-free sampling

Eight valid frames drawn at random, so order carries no meaning.

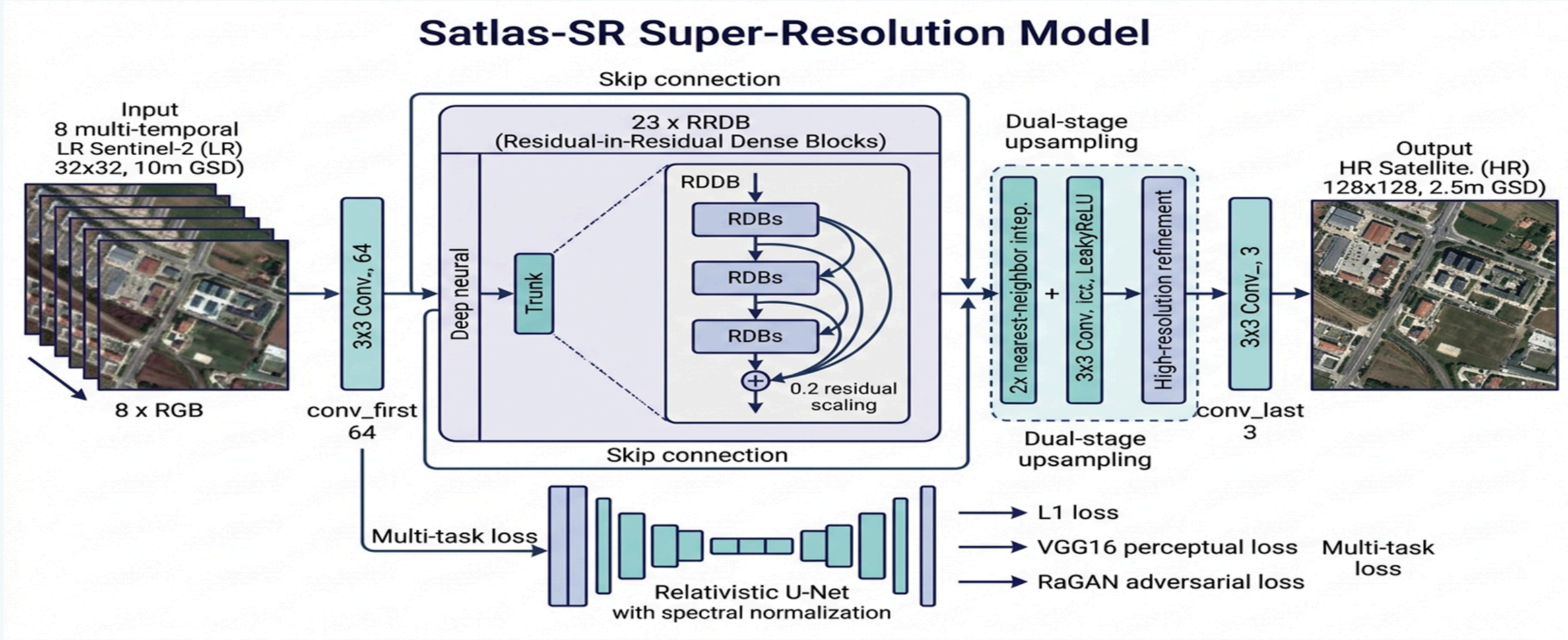
4

Normalization

Bytes divided by 255 into [0, 1] as float32.

A 23-block RRDBNet turns 32×32 into 128×128

The frozen base model: deep residual trunk, then two progressive 2× upsampling stages.



Inside the trunk: dense blocks, scaled residuals

Each Residual Dense Block (RDB) contains five convolutional layers, and each Residual-in-Residual Dense Block (RRDB) consists of three RDBs, with every layer receiving feature maps from all preceding layers within its RDB.

Layer	Input channels	Output channels
L1	64	32
L2	96	32
L3	128	32
L4	160	32
L5	192	64

Growth rate G = 32

Layers 1–4 each add 32 channels; layer 5 compresses 192 → 64.

Residual scaling α = 0.2

Scales both RDB and RRDB shortcuts so 100+ conv layers stay stable.

Progressive upsampling

Two 2× stages plus conv_hr avoid single-step 4× checkerboard artifacts.

$$xRRDB = xin + 0.2 \cdot RDB_3(RDB_2(RDB_1(xin)))$$

Understanding the data flow through our RRDBNet generator.

TRAINING OBJECTIVE

Three losses balance accuracy, structure and realism

Generator trained against a spectral-normalized U-Net discriminator; inference uses EMA weights.

L1 pixel loss

λ = 1.0

Mean absolute error vs. NAIP; avoids the blurry median colors L2 produces.

VGG-16 perceptual loss

λ = 1.0

VGG16 perceptual loss preserves the shapes, edges, and structural details of buildings and roads in the reconstructed image.

RaGAN adversarial loss

λ = 0.005

Real images must look more real than the average fake, and vice versa.

Spectral-norm U-Net discriminator



128 → 64 → 32 → 16 → 32 → 64 → 128 px

Dashed lines: skip connections. Output is a 128×128 real/fake map, one score per pixel.

$$L_G = 1.0 \cdot L_1 + 1.0 \cdot L_{percep} + 0.005 \cdot L_{RaGAN}$$

Optimizer: Adam | Learning rate: 10^{-4}

DATA PARTITION

A fair evaluation requires unseen test scenes

The dataset was audited before any scientific claim was evaluated.

Evaluation Dataset Selection and Validation

Development

107 tiles

Used to design and tune the method.

Locked holdout

107 tiles

Reserved for final evaluation.

Excluded

42

clipped or invalid

214 clean eligible tiles

Zero retraining

EMA checkpoint weights stayed frozen; no gradient updates.

No test-data tuning

Calibration statistics and blend parameters are set using development data only.

Scene-level statistics

Tests run across N = 107 independent tiles, not pixels.

Two independent signals form one uncertainty map

Spatial inconsistency and temporal sensitivity expose different reasons a prediction may fail.

SPATIAL TTA — weight 0.19

Spatial uncertainty

Prediction spread across 8 rotations and reflections.
High disagreement flags unstable regions.

0° | 90° | 180° | 270° | ⇄0° | ⇄90° | ⇄180° | ⇄270°

TEMPORAL SENSITIVITY — weight 0.81

Temporal uncertainty

Prediction spread across six temporal input variants.
High spread flags dependence on chosen frames.

S0 | S1 | S3 | S4 | S5 | S6

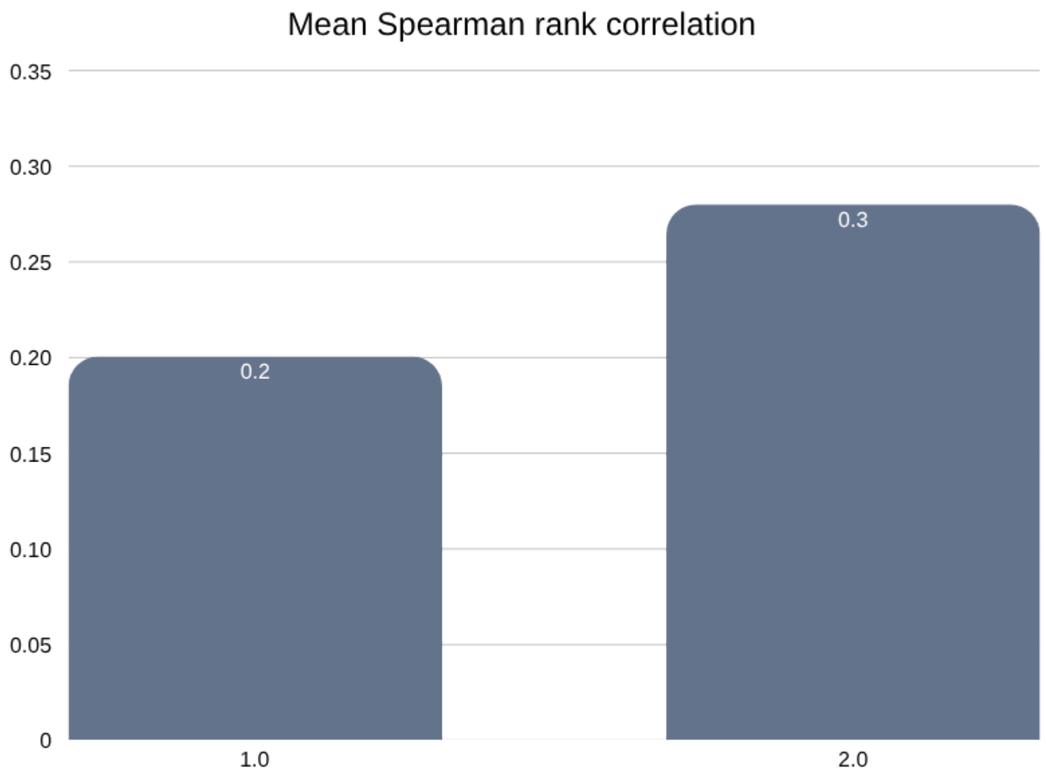
$$U(\text{hybrid}) = 0.19 \cdot Z(\,U(\text{tta})\,) + 0.81 \cdot Z(\,U(\text{input})\,)$$

Frozen on development data: spatial μ 0.0831, σ 0.0514 · temporal μ 0.0573, σ 0.0412

CALIBRATION RESULT · EXP-017

Hybrid uncertainty ranks errors 39.5% better

Locked holdout, 107 unseen geographic scenes, against NAIP ground truth.



+39.5%

Spearman rank correlation
(Estimated uncertainty,Actual reconstruction error)

+37.4%

AURG (0.00788 → 0.01083)

Area Under the Random Gain curve

93 / 107

tiles won: 86.92%

hybrid estimator performed better than the Sobel baseline on 93 out of 107 holdout tiles

RECONSTRUCTION OPERATOR

Uncertainty becomes an active correction signal

Continuous/Soft blending avoids the abrupt boundaries of hard pixel replacement.

P_o

Primary prediction

Baseline output from the primary clear acquisition S0.

Pe

Ensemble predictions

Mean of six temporal predictions; noise averages out.

w(x, y)

Soft weight field

Sigmoid of hybrid uncertainty, per pixel, in [0, 1].

Psoft

Reconstruction

Primary and ensemble mixed by the weight field.

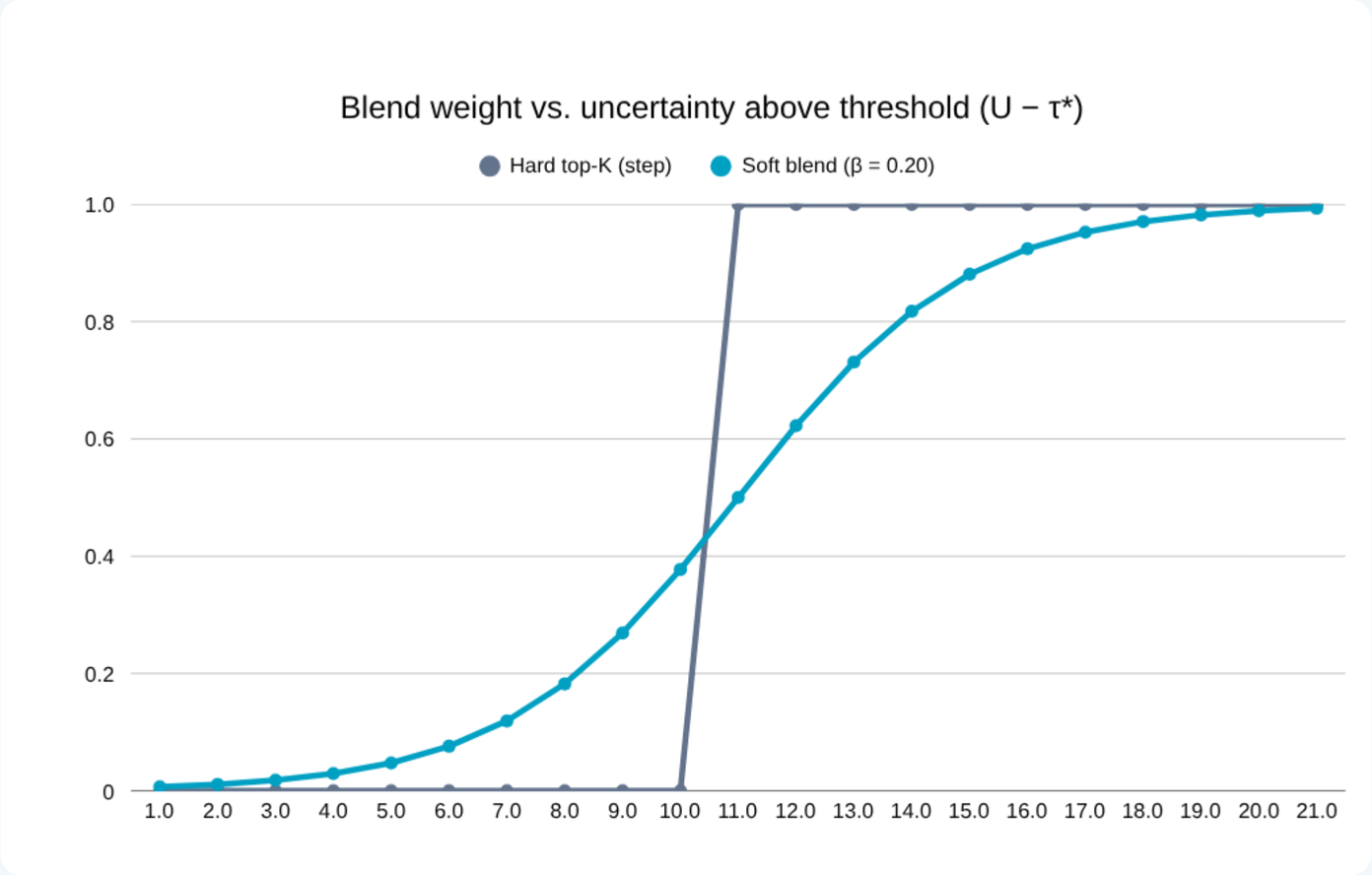
$$P_{soft} = (1 - w) \cdot P_o + w \cdot Pe \qquad w = \sigma((U(hybrid) - \tau^*) / \beta)$$

(U(hybrid) = hybrid uncertainty at that pixel, τ* = threshold calculated for the tile, β = temperature parameter(0.20)

τ* is the threshold that decides how much the model should rely on the ensemble prediction instead of the original prediction

Soft weights remove hard-replacement seams

Hard top-K replacement flips pixels from 0 to 1; the sigmoid tapers the transition.



Soft blending improves accuracy and keeps edges

Locked holdout means across 107 scenes; lower MAE and higher PSNR are better.

METHOD	MAE ↓	PSNR ↑	SOBEL GRADMAG
Baseline (primary clear acquisition)	0.08527	19.24 dB	0.0420
Hard replacement	0.08218	19.70 dB	0.0371
Continuous soft blend	**0.08146**	**19.81 dB**	**0.0362**

+0.571 dB

PSNR over baseline

96.26% of tiles improve

–0.00381

MAE reduction vs. baseline

100 / 107

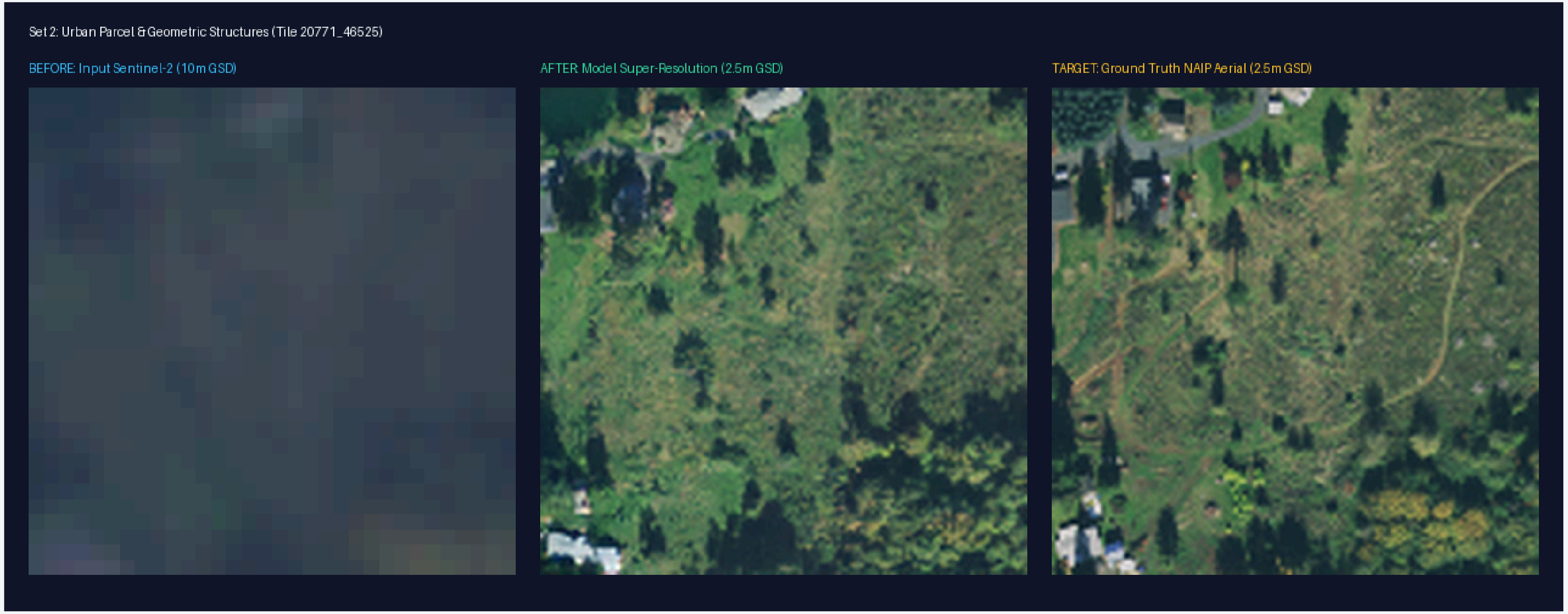
tiles where soft beats hard 20%

93.46% win rate

Qualitative Results: Super-Resolution of Transport and Urban Corridors



Qualitative Results: Super-Resolution of Transport and Urban Corridors



STATISTICAL EVIDENCE

The gains hold up statistically

Paired tests over 107 independent geographic scenes on the locked holdout (EXP-024).

−0.00072

MAE, soft vs. hard 20%

95% CI [−0.00085, −0.00059]
$p = 6.5\times10^{-20}$

100 / 7

wins vs. losses over hard 20%

No ties; Wilcoxon $p = 1.6\times10^{-17}$

d = −1.09

Cohen’s effect size

Large effect on paired MAE

+0.571 dB

PSNR over the baseline

$t = 8.56$; $p = 9.5\times10^{-14}$; 96.26% win

−20.2%

weight-field gradient $\|\nabla w\|$

Smoother transitions; $p = 6.6\times10^{-23}$

7 / 7

preregistered gates passed

Frozen preflight protocol, EXP-024

SCIENTIFIC GUARDRAILS

What the Results Mean ?

The method reduces uncertainty-related error; it cannot create information the sensor never recorded.

What ensembling can do

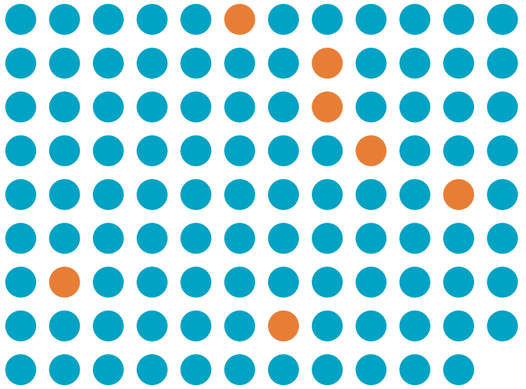
Reduce variance across temporal observations:

- Uncorrelated sensor noise
- Sub-pixel registration noise
- Transient thin-cirrus haze

It cannot recover details absent from the photons, such as power lines or rooftop antennas at 10 m.

Where it can fail

Land changed between acquisition dates: harvest, construction clearing, snowfall.



7 / 107

holdout tiles (6.54%) where ensembling was worse than baseline

IN PRODUCTION Pair with cloud masking and change detection; treat uncertainty as a review signal.

POTENTIAL IMPACT

Where more reliable high-resolution imagery can help

Potential downstream uses, not claims of a deployed operational product.

Land & vegetation

Land-use and land-cover mapping of built-up areas and vegetation.

Agriculture

Seasonal crop interpretation, paired with change masking.

Disaster & infrastructure

Inspect spatial patterns when high-resolution imagery is stale.

Geospatial analytics

Use uncertainty layers to flag regions for manual review.

Research

Reusable evaluation protocol for uncertainty-aware reconstruction.

Uncertainty layers turn “is this detail real?” into a measurable review signal.

SUMMARY

Satlas-SR makes super-resolution uncertainty-aware

The contribution is the combination, not a new backbone.

+39.5% error-rank correlation gain over the Sobel baseline

+0.571 dB PSNR gain over the published baseline

+19.8% edge sharpness vs. uniform ensembling

107 locked holdout scenes, zero retraining

TAKEAWAY Estimate uncertainty, use it to guide a smooth correction, and verify the improvement on unseen geographic scenes.

Sharper imagery is useful. Knowing where to trust it is better.

Spatial–temporal uncertainty estimation + continuous, uncertainty-guided reconstruction.

Thank you

